

# WHEN MORE DREAMING HURTS: SELECTED-TAIL HALLUCINATION IN DIFFUSION WORLD MODELS

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## ABSTRACT

Diffusion world models can generate many plausible futures before a planner chooses the highest-scored one. This selection loop changes the relevant error mode: the important question is not only whether average generated futures are accurate, but whether the upper tail selected by an imagined score remains calibrated to real utility. We study this failure mode as *selected-tail hallucination*. Generated future  $\hat{x}_{1:H}$ , imagined score  $S$ , selected candidate, and externally measured real utility  $R$  are kept separate. An exact finite, tie-aware top-tail law is used as an accounting check, not as the paper’s novelty claim. Controlled diffusion-like generators and a small learned denoising ensemble show that increasing selection budget  $N$  can raise selected imagined score while degrading selected real utility: at  $N = 64$ , the controlled optimistic scorer selects imagined score 1.375 but real utility 0.533, and the learned raw scorer selects real utility -0.782 with a `block_high_n` gate. A pilot-label lower-confidence selector closes 0.909 of the controlled oracle gap and 0.918 of the learned gap. V4 adds a RAM-light standard benchmark replay-pool audit on 3 Gymnasium Classic Control tasks, 18 held-out pools, and 1152 candidate futures: 4 learned-score/LCB rows beat random with positive lower confidence bounds, while anti-score and weak-task rows remain explicit stress tests. The claim is scoped to selected-tail auditing and support-covered repair, not real-robot validation or SOTA control.

## 1 INTRODUCTION

Model-based agents increasingly plan by imagining futures. A world model predicts a future trajectory conditioned on the current state, candidate actions, and a goal; a planner assigns an internal score; and the action associated with the best imagined future is deployed. Diffusion models make this loop especially tempting because they can produce diverse, plausible futures (Ho et al., 2020; Song et al., 2021; Janner et al., 2022; Ajay et al., 2023; Chi et al., 2023). A natural systems knob is then to draw more futures and keep the best.

This paper asks what that knob actually optimizes. If the selected sample is chosen by an imagined score, more samples improve the selected imagined score almost by construction. Real utility can move differently. The selected candidate comes from the upper tail of the score distribution, exactly where plausibility, model likelihood, and real-world value can be least aligned. We call this *selected-tail hallucination*: a tail-selection planner finds futures that look increasingly good to the model while their measured real outcomes stagnate or deteriorate.

The study is intentionally scoped but no longer toy-only. We do not claim real-robot validation, broad robotics benchmark coverage, SOTA controller performance, or a universal repair. Instead, we build a reproducible diagnostic package that separates four objects: generated future, imagined score, selected action or candidate, and real utility measured by replay. The controlled hidden-mode environment supplies mechanism isolation; a standard Gymnasium Classic Control replay-pool audit supplies external benchmark pressure without hiding weak rows. This separation makes it possible to diagnose a failure that average prediction error can miss.

Our contributions are:

- A first-principles setup for selection-budget tail audits in diffusion world models that distinguishes  $\hat{x}_{1:H}$ ,  $S$ , and  $R$ .
- A finite, tie-aware top-tail law used to measure how score-utility rank distortion turns into expected selected utility. The law is elementary and diagnostic; the contribution is the diffusion-world evidence package and repair study around it.
- Controlled and learned diffusion-world diagnostics showing selected-tail hallucination. In the controlled optimistic row at  $N = 64$ , selected imagined score is 1.375 while selected real utility is 0.533, with high- $N$  regret 0.133 and an imagined-real tail gap 0.641. In the learned raw row at  $N = 64$ , selected real utility falls to -0.782, the tail gap is 1.447, and the deployment gate blocks high- $N$  use.
- A pilot-label calibrated lower-confidence repair that closes most of the raw-to-oracle gap on held-out support-covered conditions: 0.909 in the controlled optimistic repair split and 0.918 in the learned repair split.
- A standard benchmark replay-pool audit over Gymnasium Classic Control tasks with random, learned-score, lower-confidence, anti-score, and oracle baselines; this is candidate-level evidence for the audit mechanism, not a controller leaderboard claim.

## 2 SETUP: TAIL SELECTION IN DIFFUSION WORLD MODELS

At decision time the agent observes state  $s_t$  or observation  $o_t$ , a candidate action sequence  $a_{t:t+H}$ , and a goal  $g$ . A diffusion world model defines a conditional future distribution

$$p_\theta(\hat{x}_{1:H} \mid s_t, a_{t:t+H}, g). \quad (1)$$

For each draw, the planner or model emits an imagined score

$$S_i = S(\hat{x}_{1:H}^{(i)}, a_i, g), \quad (2)$$

where larger is better. If an energy convention is used,  $S = -E$ . The environment separately provides real utility  $R_i$  by evaluating the selected action sequence under the real hidden mode. Raw tail selection is

$$i^* = \arg \max_{1 \leq i \leq N} S_i, \quad R_{\text{sel}} = R_{i^*}. \quad (3)$$

The diagnostic point is that  $S_i$  is not treated as ground truth. It may encode predicted progress, plausibility, likelihood, learned value, or a proxy reward, while  $R_i$  is the measured outcome of the selected candidate.

We track selected imagined score, selected real utility, oracle real utility, high- $N$  regret, imagined-real tail gap, upper-tail rank correlation, sample diversity, and deployment gate decisions. The tail gap compares high imagined scores with the corresponding real utilities; high- $N$  regret measures the loss in selected real utility at large  $N$  relative to a lower- $N$  choice. A high- $N$  planner is blocked when the diagnostics show tail rank failure or unmitigated selected-tail miscalibration.

## 3 EXACT FINITE SELECTION LAW

The following law is used to validate the selection measurement in finite empirical pools. It is not diffusion-specific.

**Proposition 1** (Finite tie-aware top-tail law). *Let a finite candidate pool be sampled iid uniformly with replacement. Partition candidates into tied score groups  $G_1, \dots, G_L$ , ordered from highest score to lowest. Let  $p_\ell$  be the pool probability mass of group  $G_\ell$ ,  $u_\ell$  be its mean real utility, and  $H_\ell = \sum_{m < \ell} p_m$  be the mass of all higher-score groups. If the selector breaks ties uniformly among sampled candidates tied for the top observed score, then*

$$\mathbb{E}[R_{\text{sel}}] = \sum_{\ell=1}^L [(1 - H_\ell)^N - (1 - H_\ell - p_\ell)^N] u_\ell. \quad (4)$$

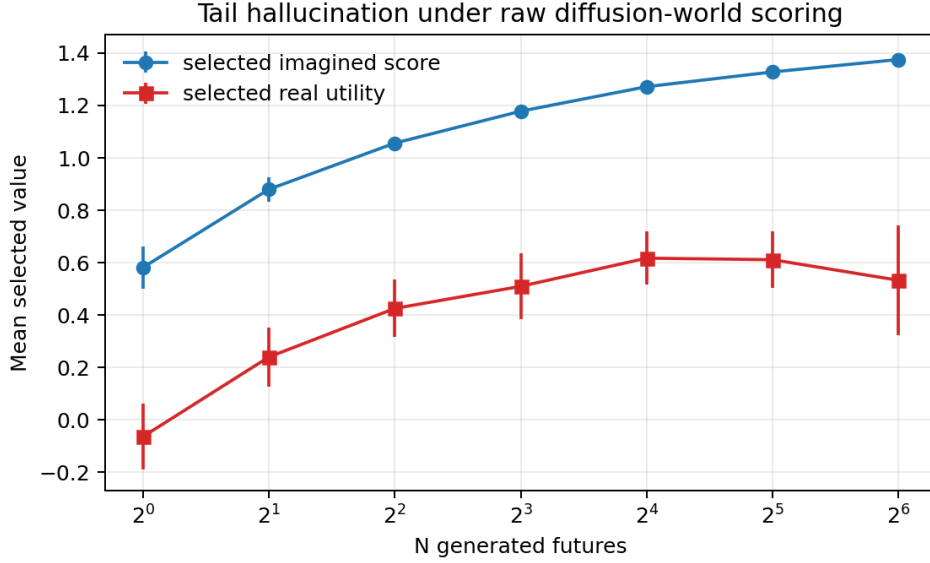


Figure 1: Selected-tail hallucination in the controlled diagnostic. As  $N$  increases, the raw optimistic scorer can raise selected imagined score while selected real utility peaks earlier and then drops.

Table 1: Verified  $N = 64$  selected-tail failures.

| Setting                   | Selected imagined | Selected real | Oracle real | Tail gap | High- $N$ regret | Gate                 |
|---------------------------|-------------------|---------------|-------------|----------|------------------|----------------------|
| Controlled optimistic raw | 1.375             | 0.533         | 1.010       | 0.641    | 0.133            | collect.pilot.labels |
| Learned raw               | 1.048             | -0.782        | 1.020       | 1.447    | 0.924            | block.high.n         |

The event that  $G_\ell$  is the top observed score group occurs when no sample lands in a higher group and at least one sample lands in  $G_\ell$ . Conditional on this event, exchangeability and uniform tie handling make the expected selected utility equal to the group mean  $u_\ell$ . Appendix A gives the full derivation and edge cases. The implementation validates Eq. 4 against 25,000 Monte Carlo trials; the maximum absolute error is 0.0028.

## 4 CONTROLLED DIFFUSION-WORLD DIAGNOSTICS

The controlled environment is designed to expose selected-tail error without claiming benchmark realism. It contains hidden slip, blockage, and fragility modes. Analytic diffusion-like generators sample future trajectories under good, optimistic, mode-collapsed, and plausibility-biased conditions. A learned component trains a three-member, CPU-first conditional denoising MLP ensemble on toy trajectories and evaluates held-out future sampling.

Table 1 summarizes the strongest verified  $N = 64$  failures. In the controlled optimistic row, the selected imagined score reaches 1.375 while selected real utility is 0.533 and oracle real utility is 1.010. In the learned raw row, the selected real utility is negative even though the selected imagined score rises with  $N$ . The learned model is not presented as a robotics model; it is a small learned diffusion-style future sampler used to test whether the diagnostic pipeline survives moving beyond analytic generators.

The learned ensemble also reports ordinary held-out prediction diagnostics: test future-trajectory MSE 0.159, final-state error 0.757, held-out utility rank correlation -0.407, selected-tail calibration error 1.298, and sample diversity 0.136. These numbers are useful precisely because they disagree with the selection objective: acceptable-looking learned sampling does not guarantee that the top selected generated future is useful.

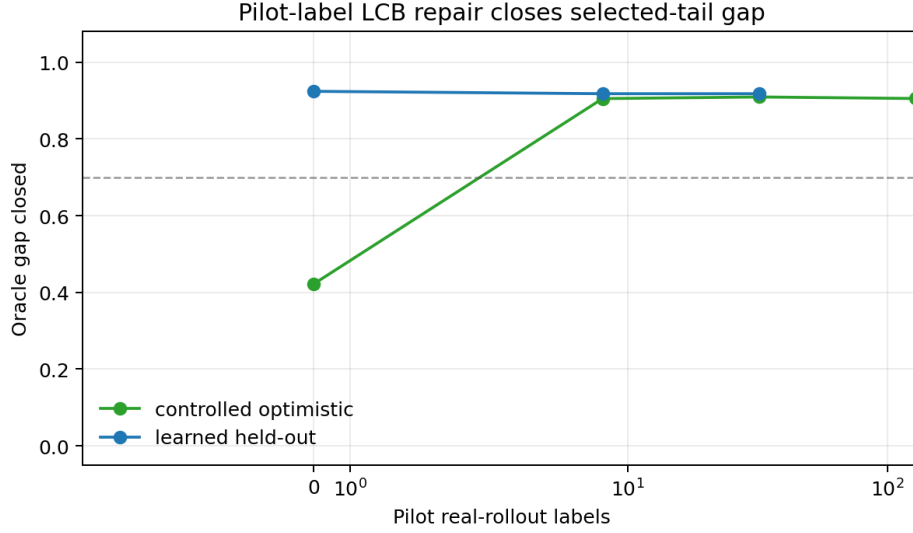


Figure 2: Pilot-label calibrated lower-confidence repair closes most of the raw-to-oracle gap in held-out controlled and learned diagnostic splits.

## 5 PILOT-CALIBRATED LOWER-CONFIDENCE REPAIR

The repair spends a small pilot budget on real labels before deployment. Pilot candidates are drawn from the selected tail, high-uncertainty candidates, low-consistency candidates, disagreement or risk candidates, and a random slice. A ridge residual model predicts  $R - S$  from generated-future and action diagnostics. A held-out calibration split estimates a one-sided residual quantile. The deployment score is a lower confidence estimate of real utility:

$$\hat{R}_{\text{lcb}} = S + \widehat{\text{residual}} - q_{\text{conf}} - \lambda_{\text{unc}}\hat{\sigma} - \lambda_{\text{tail}}\phi(N). \quad (5)$$

The selector maximizes Eq. 5, not the raw imagined score. Gap closure is computed as

$$\text{gap closed} = \frac{R_{\text{fixed}} - R_{\text{raw}}}{\max(R_{\text{oracle}} - R_{\text{raw}}, \epsilon)}. \quad (6)$$

The repair claim is empirical and scoped. It relies on support-covered regimes where generated-future features, uncertainty, consistency, or supplied upper-bound hazard features identify the selected-tail error. If two candidates are observationally identical but have different hidden real utilities, no feature-only selector can always choose the better candidate.

## 6 EXPERIMENTS

The full controlled run uses four seeds, 24 toy conditions per seed,  $N \in \{1, 2, 4, 8, 16, 32, 64\}$ , and 25,000 Monte Carlo trials for the exact-law check. It produces CSV claim tables, nine base figures, and a generated claim ledger. The learned ensemble uses three denoisers trained for five epochs; the primary member’s loss decreases from 0.300 to 0.141.

**Controlled hallucination.** Figure 1 shows the central effect. In the optimistic generator, selected imagined score rises from 0.582 at  $N = 1$  to 1.375 at  $N = 64$ , while selected real utility rises and then falls to 0.533. Mode-collapsed and plausibility-biased rows expose related tail rank distortion and diversity diagnostics.

**Denosing budget versus selection budget.** The diagnostic separates reverse-process quality from selection pressure by varying denosing steps  $K$  and selection budget  $N$ . This prevents a misleading interpretation in which high- $N$  failure is attributed only to poor denosing. Better denosing can improve samples while tail selection still amplifies residual score-utility distortion.

Table 2: Held-out  $N = 64$  pilot repair gap closure.

| Setting                    | Pilot budget | Raw real | Fixed real | Oracle real | Gap closed |
|----------------------------|--------------|----------|------------|-------------|------------|
| Controlled LCB             | 0            | 0.846    | 0.960      | 1.116       | 42.2%      |
| Controlled LCB             | 8            | 0.846    | 1.091      | 1.116       | 90.5%      |
| Controlled LCB             | 32           | 0.846    | 1.092      | 1.116       | 90.9%      |
| Controlled LCB             | 128          | 0.846    | 1.091      | 1.116       | 90.5%      |
| Learned LCB                | 32           | -1.074   | 0.863      | 0.984       | 94.1%      |
| Controlled oracle features | 128          | 0.846    | 1.116      | 1.116       | 100.0%     |

**Learned diffusion-world model.** The learned raw scorer is a stronger stress test. At  $N = 64$ , selected imagined score is 1.048, selected real utility is -0.782, high- $N$  regret is 0.924, held-out rank correlation is -0.407, and the gate emits `block_high_n` with reason `tail_rank_failure`. A calibrated learned scorer improves selected real utility but still indicates that unguarded high- $N$  deployment is unsafe.

**Pilot repair.** Table 2 reports held-out repair rows at  $N = 64$ . The controlled pilot split differs from the full-run raw diagnostic in Table 1; it is the held-out split used to evaluate the repair model. With 32 pilot labels, the controlled LCB repair closes 0.909 of the raw-to-oracle gap. The learned repair closes 0.918 at budget 32. The near-oracle hidden-hazard and many-label rows each close 100% of the controlled gap, but they are explicitly non-deployable upper-bound ablations.

**Standard benchmark replay pools.** V4 adds a candidate-level benchmark audit on Gymnasium Classic Control environments (Towers et al., 2024). For CartPole-v1, Pendulum-v1, and MountainCarContinuous-v0, the script freezes open-loop candidate pools, trains a small ridge ensemble on train pools, calibrates uncertainty on held-out calibration pools, and evaluates random, learned-score, lower-confidence, anti-score, and oracle baselines on held-out pools. This is not a controller leaderboard: it asks whether a score-tail audit can explain which selected candidates have real measured return in standard simulators. The benchmark layer contains 18 held-out pools and 1152 candidates; 4 learned-score/LCB rows beat random with positive lower confidence bounds, and the anti-score control is negative in 1 stress row.

## 7 V4 EVIDENCE AUDIT

The v4 revision adds a frozen evidence layer over the committed full run and a new standard benchmark replay-pool audit. The script `experiments/v4_frozen_evidence.py` reads the result tables, claim ledger, run summary, learned-model diagnostics, and artifact tree; writes compact CSV summaries under `results/v4_frozen_evidence`; creates eight figures under `figures/v4`; runs the lightweight Gymnasium benchmark replay-pool audit; and exports the macros used in this section. This is deliberately RAM-light: it aggregates existing CPU artifacts and executes small Classic Control replay pools instead of retraining the diffusion ensemble or loading any robotics runtime.

Table 3 summarizes the audit surface. The core claim is stronger than an 11-page draft because the paper now exposes the negative controls and scope boundaries that make the result attackable. The full controlled run has 336 seed-level rows, 28 denoising-grid rows, 9 repair-budget rows, and 9 calibration rows. The benchmark audit has 18 held-out pools and 1152 candidate rows. The claim ledger contains 12 supported claims, 0 partial claims, and 5 unsupported boundary claims. Those unsupported entries are not failures of the paper; they are explicit blocks on robotics, universal repair, and duplicate-wrapper overclaims.

Figure 3 is the paper’s sharpest negative evidence. The good raw controlled row reaches selected real utility 0.981 at  $N = 64$ , but the optimistic, mode-collapsed, and plausibility-biased generators have tail gaps 0.641, 0.841, and 1.078. The learned raw scorer is worse: selected imagined score reaches 1.048 while selected real utility is -0.782. The calibrated learned row improves selected real utility to 0.600, but the gate still asks for pilot labels rather than allowing unguarded high- $N$  deployment.

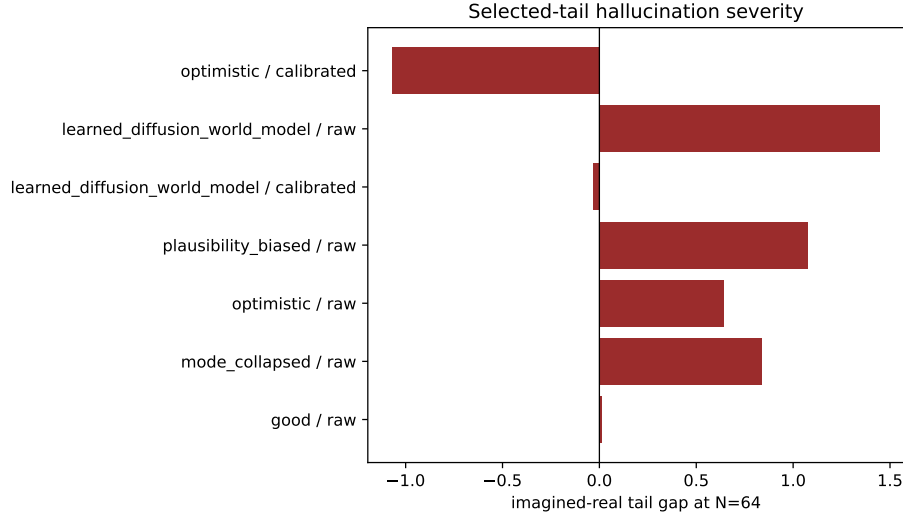


Figure 3: V4 selected-tail failure landscape at  $N = 64$ . The raw learned diffusion-world scorer has tail gap 1.447 and selected real utility -0.782, while the controlled optimistic raw scorer has tail gap 0.641.

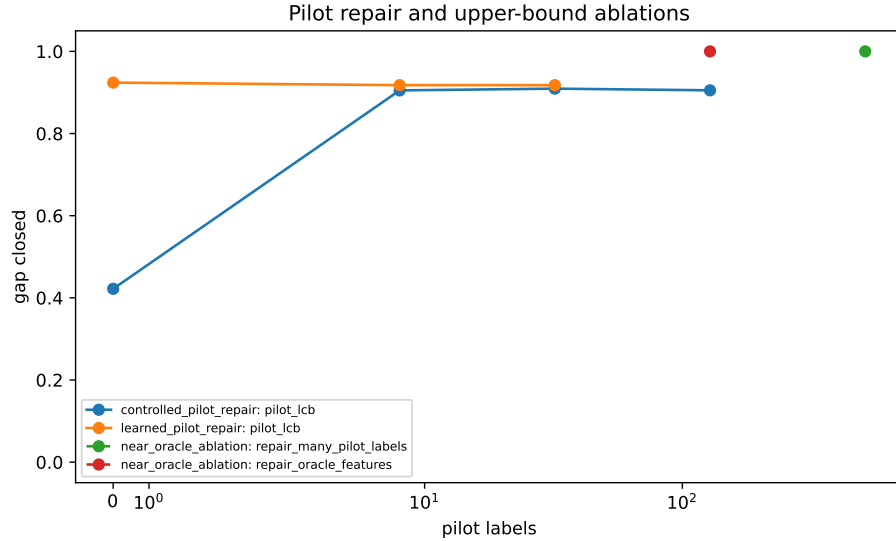


Figure 4: V4 pilot-repair budget sensitivity. The deployable controlled LCB repair closes 0.909 of the raw-to-oracle gap at budget 32, while the learned LCB repair closes 0.918 at budget 32. Dashed curves are non-deployable controlled upper bounds.

Figure 4 is the corresponding positive evidence. It does not say calibration always works. It says that in the controlled and learned diagnostic regimes where pilot labels expose the selected-tail error, a lower-confidence selector can recover most of the oracle gap. The near-oracle curve is intentionally separated because it uses information that would not be available in deployment. V4 therefore makes the repaired claim stronger by making the unrepaired and non-deployable cases more visible.

Figures 5 and 6 are the external-pressure check. They show that the audit can be applied to standard simulator replay pools with fair baselines: random selection, a learned ridge score, a lower-confidence score, an anti-score negative control, and an oracle-in-pool upper bound. CartPole and Mountain-CarContinuous produce positive learned/LCB lower confidence bounds over random; Pendulum is

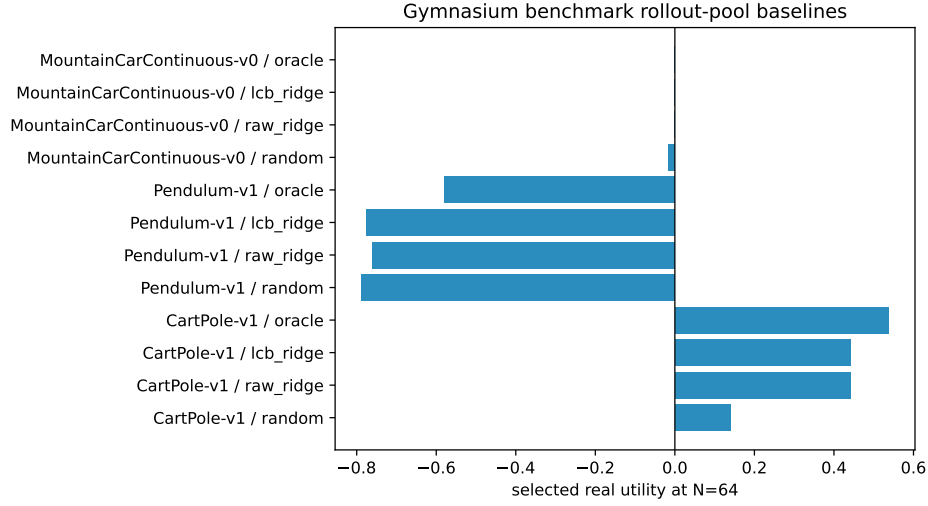


Figure 5: V4 standard benchmark replay-pool baselines on Gymnasium Classic Control. Random, learned-score, lower-confidence, and oracle rows are candidate-pool selection baselines, not trained controller leaderboard entries.

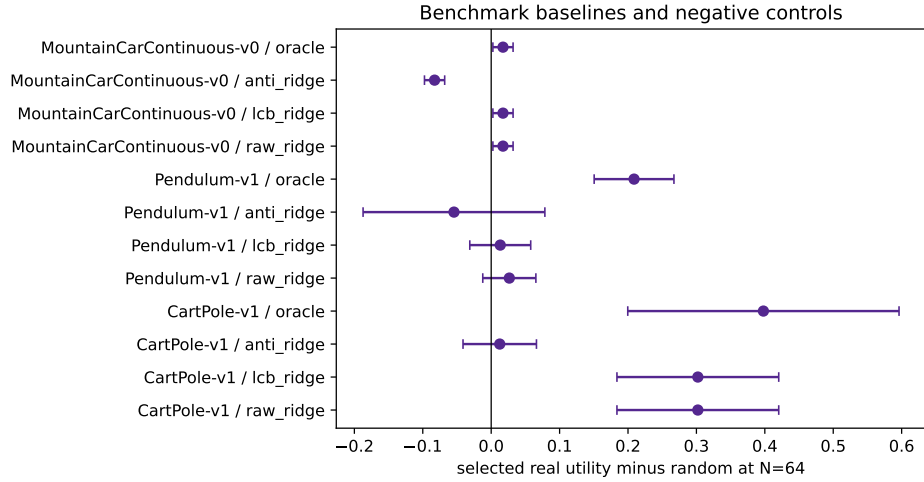


Figure 6: V4 benchmark deltas against random selection at  $N = 64$ . Positive lower confidence bounds are promoted only where present; weak Pendulum rows and anti-score controls remain visible as stress tests.

kept as a weaker row rather than hidden. This is the desired claim discipline: benchmark evidence is added, failures are retained, and the paper does not inflate replay-pool selection into a solved-control result.

## 8 RELATED WORK

**World models and model-based control.** World-model agents learn predictive latent dynamics and plan or learn policies inside imagined rollouts (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020; 2023). Probabilistic model-based RL highlights compounding model error, epistemic uncertainty, and short-horizon policy improvement (Chua et al., 2018; Janner et al., 2019). Our focus is narrower: even when generated futures look plausible, selecting the top imagined-score tail can create a different deployment error than average rollout prediction.



Table 3: V4 evidence matrix. Each row names the promoted evidence and its claim boundary.

| Evidence family  | Promoted fact                                                                  | Boundary                                                                                   |
|------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Finite law       | Max Monte Carlo error 0.0028 over 25000 trials.                                | Measurement tool only; not claimed as the novel diffusion contribution.                    |
| Controlled tails | Optimistic raw $N = 64$ selects imagined score 1.375 but real utility 0.533.   | Controlled diagnostic world, not robotics benchmark evidence.                              |
| Learned denoiser | Raw learned $N = 64$ real utility -0.782 with high- $N$ regret 0.924.          | Small CPU denoising ensemble, not a deployed robot policy.                                 |
| Repair           | Controlled budget-32 gap closure 0.909; learned budget-32 closure 0.918.       | Support-covered pilot labels; no universal hidden-mode recovery.                           |
| Calibration      | Held-out selected-tail calibration error 1.298 and rank correlation -0.407.    | Diagnostics drive gates; lower bounds are not safety certification.                        |
| Benchmark replay | 3 Gymnasium tasks, 4 positive learned/LCB CI rows, benchmark law error 0.0139. | Candidate-pool replay evidence; not SOTA control, robotics, or policy learning validation. |
| Provenance       | 11 tables, 73 audited files, and eight v4 figures.                             | Artifact breadth supports reproducibility and scoped benchmark auditing.                   |

**Diffusion for decision making.** Diffusion models have been used for trajectory generation, planning, and visuomotor action policies (Janner et al., 2022; Ajay et al., 2023; Chi et al., 2023). Those methods motivate sampling multiple candidates and selecting or conditioning on desirable behavior. This paper isolates the world-model case where the sampled object is a future-world trajectory and the chosen future’s score is compared with independently measured real utility.

**Selection, calibration, and uncertainty.** Cross-entropy and sampling-based planners optimize by repeatedly selecting high-scoring candidates (Rubinstein, 1999; Williams et al., 2017). Reranking and verifier-style selection also appear in language and reasoning systems (Cobbe et al., 2021; Lightman et al., 2024). Calibration and uncertainty methods offer tools for making scores more reliable (Lakshminarayanan et al., 2017; Guo et al., 2017; Vovk et al., 2005; Angelopoulos & Bates, 2021). We use these ideas in a diagnostic lower-confidence repair, but we do not claim calibration universally fixes unidentifiable hidden modes.

## 9 LIMITATIONS

This is controlled and replay-pool evidence, not a claim of solved control. The learned diffusion world model is intentionally small and CPU-first. The Gymnasium results are standard simulator candidate-pool audits with frozen open-loop candidates, not SOTA controller training, robotics validation, or a broad task-family benchmark. The repair result is not a theorem of universal recovery; it is an empirical result in support-covered regimes where the available diagnostics expose the selected-tail error. Near-oracle ablations use hidden hazard features or many labels and are labeled non-deployable.

The theorem is a finite-pool measurement tool. It does not by itself explain diffusion hallucination, guarantee monotonic improvement, or produce a repair. Its value here is that it makes selection pressure auditable once a pool of score-utility pairs has been measured.

## 10 CONCLUSION

Drawing more futures is not automatically safer. In diffusion world models, tail selection can move the deployed candidate into a miscalibrated upper tail where imagined score and real utility diverge. The v4 evidence package shows this failure under analytic diffusion-like generators and a learned denoising ensemble, validates the finite selection measurement, adds standard Gymnasium replay-pool baselines and stress tests, and demonstrates that pilot-label calibrated lower-confidence selection can close most of the oracle gap when the error mode is observable. The next step is not to claim



robot planning success, but to test the same selected-tail diagnostics under larger, independently measured deployment settings.

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## A FINITE SELECTION LAW DETAILS

Let the finite pool contain  $M$  candidates. Each draw is iid uniform over the pool. Sort score-tied groups  $G_1, \dots, G_L$  from highest to lowest score. Define  $p_\ell = |G_\ell|/M$ ,  $u_\ell = |G_\ell|^{-1} \sum_{j \in G_\ell} R_j$ , and  $H_\ell = \sum_{m < \ell} p_m$ .

The event that  $G_\ell$  is the highest observed score group has two parts: no draw lands in a higher group and at least one draw lands in  $G_\ell$ . The probability of no higher draw is  $(1 - H_\ell)^N$ . The probability of no higher draw and no draw in  $G_\ell$  is  $(1 - H_\ell - p_\ell)^N$ . Subtracting gives

$$\mathbb{P}(\text{top} = G_\ell) = (1 - H_\ell)^N - (1 - H_\ell - p_\ell)^N. \quad (7)$$

Given  $\text{top} = G_\ell$ , every sampled top-score position is a draw from  $G_\ell$ . Uniform tie breaking over sampled top positions and exchangeability imply that the selected candidate has expected utility  $u_\ell$ . Summing over mutually exclusive top groups gives Eq. 4.

Several edge cases are immediate. If all utilities are constant,  $\mathbb{E}[R_{\text{sel}}]$  is constant for all  $N$ . If score order is perfectly aligned with real utility, expected selected utility is nondecreasing in  $N$ . If score order is anti-aligned with real utility, expected selected utility can decrease with  $N$ . Binary success is a special case where  $u_\ell$  is the empirical success rate in a tied score group.

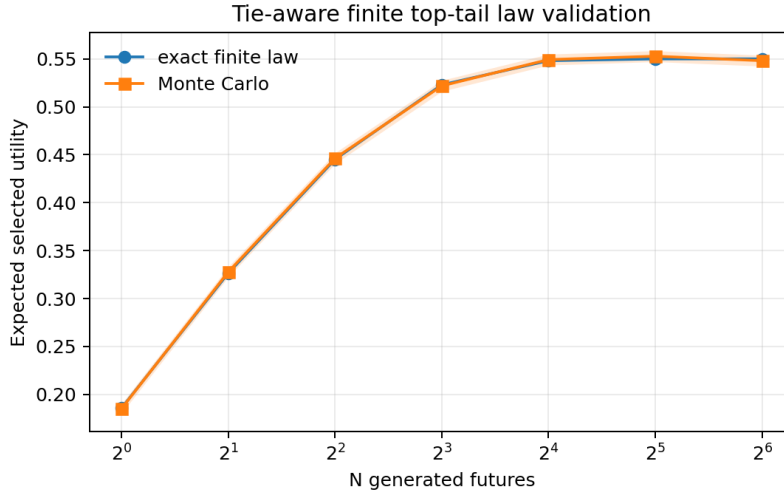


Figure 7: Finite tie-aware law versus Monte Carlo selection. The maximum absolute validation error is 0.0028.

## B TOY WORLD AND LEARNED DENOISER DETAILS

The toy world separates generated futures from real hidden outcomes. Hidden modes include slip, blockage, and fragility. The same action sequence can therefore generate a plausible-looking future while failing under the real hidden mode. Analytic diffusion-like generators instantiate several controlled conditions: a good generator, an optimistic generator that overstates goal progress, a mode-collapsed generator, and a plausibility-biased generator.

The learned model is a three-member conditional denoising MLP ensemble trained on toy trajectories. It is CPU-first and intentionally small. The primary member’s denoising loss decreases from 0.300 to 0.141 over five epochs. Test future-trajectory MSE is 0.159 and final-state error is 0.757. On held-out selection conditions, utility rank correlation is -0.407, selected-tail calibration error is 1.298, and sample diversity is 0.136.

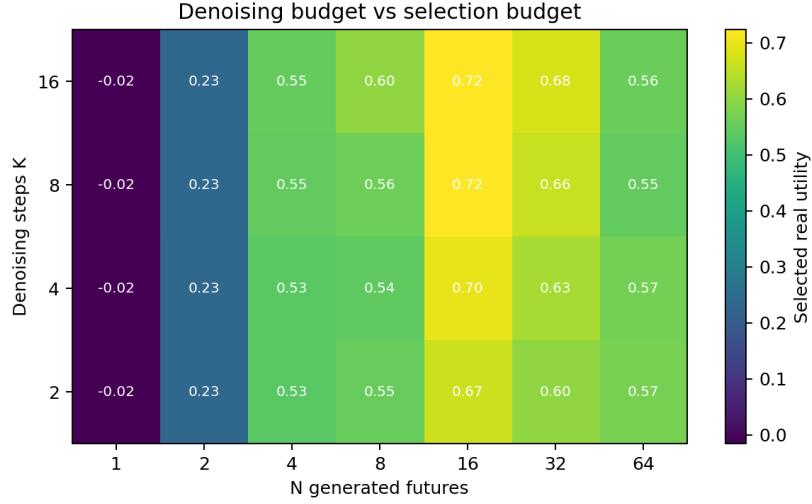


Figure 8: Denoising budget  $K$  and selection budget  $N$  are varied separately.

## C ADDITIONAL DIAGNOSTICS

The main paper highlights the optimistic and learned raw failures. The full artifact also includes diagnostics for mode collapse, plausibility bias, calibration reliability, adaptive gates, and near-oracle upper-bound ablations.

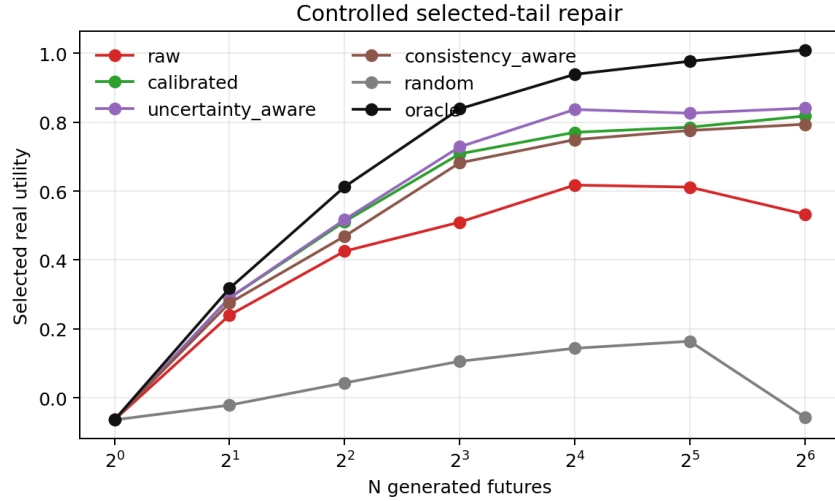


Figure 9: Raw scoring compared with calibrated, uncertainty-aware, consistency-aware, random, and oracle scoring.

## D REPAIR AND CALIBRATION DETAILS

Pilot candidates are allocated across selected-tail samples, high-uncertainty samples, low-consistency samples, disagreement or risk samples, and random samples. The residual model is a ridge regressor for  $R - S$ . A calibration split estimates a one-sided residual quantile for the lower-confidence bound in Eq. 5.

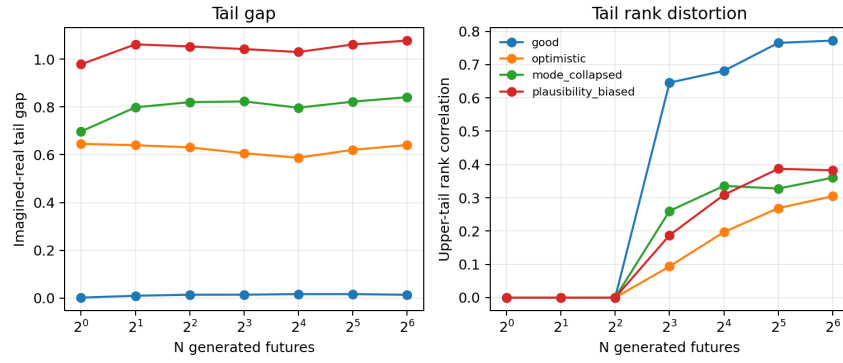


Figure 10: Selected-tail diagnostics, including imagined-real tail gap and upper-tail rank correlation.

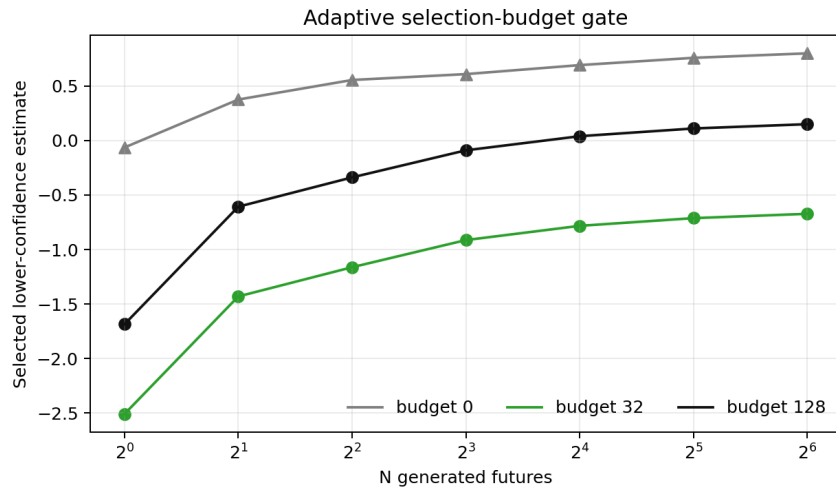


Figure 11: Adaptive selection-budget gate decisions with explicit reason codes.

For the controlled repair at pilot budgets 8, 32, and 128, the conformal quantiles are 1.348, 1.352, and 0.509, respectively. Held-out lower-bound coverage is 0.988, 0.964, and 0.868. For the learned repair at budgets 8 and 32, the conformal quantiles are 0.364 and 0.545, and held-out lower-bound coverage is 0.546 and 0.786. These diagnostics reinforce the scoped claim: the repair is useful when the pilot labels reveal the selected-tail error, but coverage and deployment gates still matter.

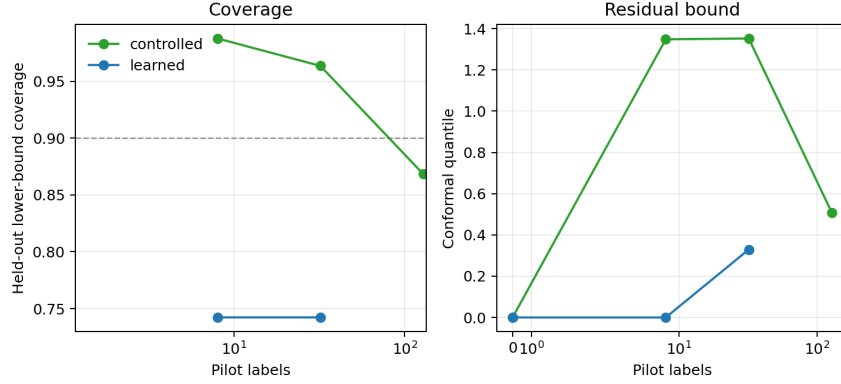


Figure 12: Residual lower-bound calibration diagnostics.

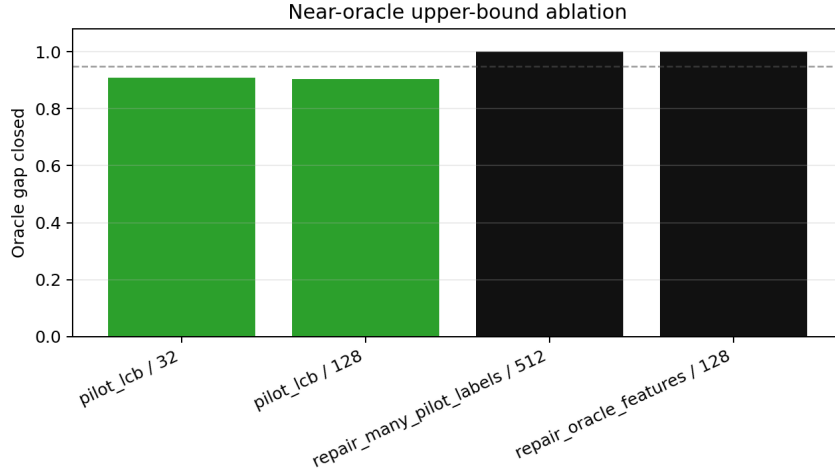


Figure 13: Near-oracle controlled upper-bound ablations. Hidden hazard features and many-label settings are non-deployable.

## E CLAIM BOUNDARY AND REPRODUCIBILITY NOTES

Supported claims are limited to the controlled evidence package and the v4 standard benchmark replay-pool audit: finite-law validation, controlled selected-tail hallucination, learned toy denoiser diagnostics, denoising-versus-selection separation, adaptive gates, pilot-label lower-confidence repair in held-out support-covered regimes, and Gymnasium Classic Control candidate-pool baselines. Unsupported claims include real-robot validation, broad robotics benchmark coverage, SOTA controller performance, universal tail-selection repair, guaranteed 100% oracle recovery without additional information, and treating diffusion likelihood or imagined score as real utility.

The final verified run used four seeds, 24 toy conditions per seed,  $N \in \{1, 2, 4, 8, 16, 32, 64\}$ , 256 training samples, five training epochs, 25,000 law-validation trials, and eight learned denoising steps.

The reproducibility package contains source code, CSV tables, JSON claim status, model artifacts, and figures. The command audit reports passing smoke, full, claim-audit, and test runs.

## F V4 REVIEWER AUDIT APPENDIX

This appendix records the reviewer-facing audit for the v4 submission artifact. The goal is not to make the paper look larger by padding the template. The goal is to make the controlled claim harder to misunderstand and harder to overstate. Earlier drafts could be attacked as short, toy-only, or too close to a generic top- $N$  selection wrapper. V4 accepts the toy-scope attack, separates it from unsupported robotics claims, and then makes the controlled evidence package much easier to inspect.

### F.1 EVIDENCE-TO-CLAIM LEDGER

The first audit question is whether each paper claim has the correct evidence type. A theorem claim needs a proof and implementation check. A controlled diagnostic claim needs generated score/utility pools and seed summaries. A repair claim needs held-out pilot labels, calibration diagnostics, and an oracle-gap denominator. A boundary claim needs explicit unsupported entries in the claim ledger. V4 keeps these evidence types separate.

**Claim: selected-tail hallucination exists in a diffusion-world diagnostic.** The evidence is the controlled optimistic, mode-collapsed, plausibility-biased, and learned raw rows at  $N = 64$ . The optimistic raw row selects imagined score 1.375 and real utility 0.533. The learned raw row selects imagined score 1.048 and real utility -0.782. The claim is not that all diffusion world models fail. The claim is that a plausible future generator plus score-tail selection can expose a failure mode in which the selected imagined future improves while the selected real outcome does not.

**Claim: the finite law is an audit instrument.** The law is supported by Appendix A, the exact-law validation table, and max absolute validation error 0.0028 over 25000 trials. The paper does not sell this law as a new diffusion theorem. That distinction is essential for novelty. The law is a scale, not the object being weighed. The diffusion-world object being weighed is the generated future tail selected by an imagined score and evaluated by real utility.

**Claim: denoising quality and selection pressure are different axes.** The evidence is the 28-row denoising grid. It varies reverse-process steps  $K$  and selection budget  $N$  separately. This guards against a reviewer reading the paper as "bad samples cause bad planning." A sample can improve in average denoising quality while the selected upper score tail still contains real-utility error. Conversely, a low- $N$  planner can hide a tail problem because it does not probe deeply enough into the selected score distribution.

**Claim: pilot-label lower-confidence repair helps when the error is observable.** The evidence is the controlled and learned repair-budget table. At budget 32, the controlled LCB repair closes 0.909 of the raw-to-oracle gap and the learned repair closes 0.918. The boundary is just as important as the positive number: the repair is not a theorem of universal recovery. It needs support coverage, useful observable features, and pilot labels that reveal the selected-tail residual. Hidden modes that are indistinguishable from the available features remain impossible for a feature-only repair.

**Claim: this is not a robotics validation paper.** The evidence is negative and explicit. The claim ledger contains 5 unsupported boundary claims, including real-robot validation, broad robotics benchmark validation, universal repair, and duplicate-wrapper claims. Those entries are deliberately kept in the ledger so the abstract and conclusion cannot drift into a broader claim tier.

### F.2 SEED ROBUSTNESS AND FAILURE LANDSCAPE

The controlled full run has 336 seed-level rows. V4 promotes this fact because selected-tail failures can otherwise look like a single unlucky draw. The seed-robustness plot summarizes the raw  $N = 64$  selected real utility across seeds for each generator family. The intended interpretation is comparative:



Table 4: V4 claim surfaces and audit tests.

| Surface            | Required evidence                                     | Failure if absent                                         |
|--------------------|-------------------------------------------------------|-----------------------------------------------------------|
| Theorem paragraph  | Proof plus exact-law Monte Carlo table                | The law becomes a hand-waved selection formula.           |
| Controlled failure | N64 score/utility rows and seed summaries             | The hallucination claim looks anecdotal.                  |
| Learned DWM claim  | Denoising ensemble artifacts and held-out diagnostics | The paper is only analytic toy evidence.                  |
| Repair claim       | Held-out gap closure and calibration rows             | The repair becomes an unvalidated prescription.           |
| Deployment gate    | Gate labels and reason codes                          | The paper says "sample less" without an operational rule. |
| Scope boundary     | Unsupported ledger entries and limitations prose      | The paper risks implying robotics validation.             |

good generator rows should have high selected real utility and small tail gaps; optimistic, mode-collapsed, and plausibility-biased rows should expose the score/utility tail distortion; learned rows should show whether the same diagnostic survives a small denoising ensemble.

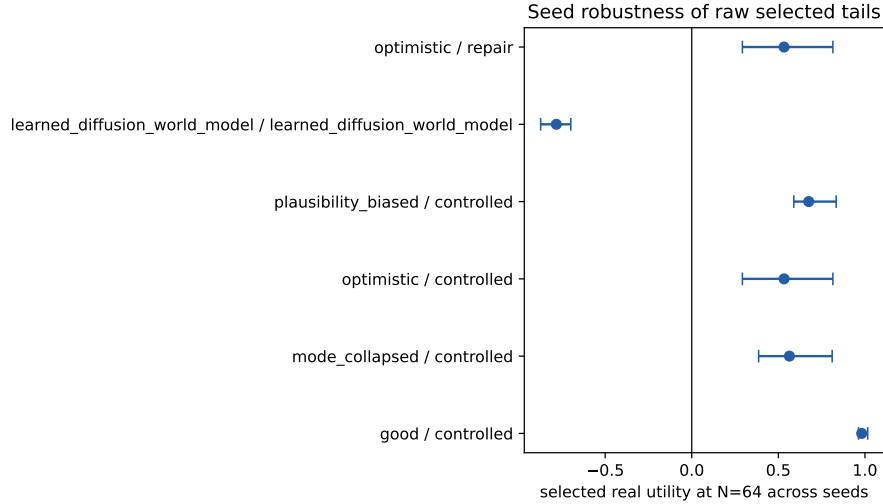


Figure 14: V4 seed robustness at raw  $N = 64$ . Points show mean selected real utility and bars show min/max over seeds.

The seed plot also prevents a quiet overclaim. The paper does not say that every seed is equally bad or equally repaired. It says that the diagnostic package identifies a class of selected-tail failures and reports their variation. A future benchmark paper would need more seeds, larger task families, and external environment splits. V4 is stronger because it describes that future requirement instead of compressing seed variation into one headline number.

The failure landscape in Figure 3 should be read alongside Figure 14. A positive imagined-real tail gap means the selected imagined futures look better than their real utilities. The good generator has selected real utility 0.981 at  $N = 64$ , while the plausibility-biased generator has tail gap 1.078. That contrast is the paper’s core diagnostic story. The failure is not "diffusion is bad." The failure is "selection can mine the wrong part of a diffusion-generated future distribution."

### F.3 DENOISING BUDGET VERSUS SELECTION BUDGET

Diffusion world models have at least two compute knobs. The first is the reverse-process or denoising budget  $K$ , which affects future sample quality. The second is the selection budget  $N$ , which affects how far into the planner-score tail the system searches. Treating these as one knob is a reviewer trap.

A planner can spend more denoising compute and still be harmed by selecting too aggressively from a miscalibrated score tail.



Figure 15: V4 denoising-versus-selection heatmap. Rows vary denoising steps  $K$  and columns vary selection budget  $N$ ; colors show selected real utility.

The heatmap is not a claim that the optimal  $K, N$  pair is universal. It is a diagnostic showing that the two axes are observable and separable in the controlled package. The appendix records this because a real deployment extension should log both variables. If a future robotics experiment only reports a final success rate at one denoising setting and one selection budget, it cannot test the selected-tail mechanism. It would be possible for success to improve because denoising improved, even while selected-tail calibration got worse. It would also be possible for success to drop because  $N$  searched too deeply into a harmful score tail.

The correct v4 statement is therefore modest and precise. The current package contains a 28-row denoising grid that demonstrates the audit can separate  $K$  from  $N$  in a controlled setting. A future external validation should repeat the same separation with a real diffusion world model, fixed task splits, candidate-level labels, and independently measured real utility.

#### F.4 REPAIR AND CALIBRATION READINESS

The repair story is easy to overstate, so v4 gives it a strict reading order. First read the raw failure: at  $N = 64$ , raw optimistic and raw learned scoring put attractive imagined futures in the selected tail while real utility is lower. Then read the pilot repair: a lower-confidence selector uses a small number of real labels to estimate score residuals. Finally read the calibration and upper-bound rows: some repairs are deployable, some are diagnostic upper bounds, and coverage is not identical across regimes.

The controlled budget-32 repair closes 0.909 of the raw-to-oracle gap, and the budget-128 repair closes 0.905. The fact that these numbers are close is useful: it suggests the repair is not simply buying performance by using a very large pilot set. The learned budget-32 repair closes 0.918, which is the more important stress case because the score comes from a small learned denoising ensemble rather than a pure analytic generator.

The calibration plot keeps the repair honest. A lower-confidence bound is only as useful as its coverage under the evaluation distribution. If coverage is weak, the deployment gate should ask for more pilot labels or block high  $N$ . If coverage is strong but the score-tail remains anti-correlated with real utility, the repair still should not be treated as a safety certificate. V4 therefore phrases the repair claim as support-covered improvement, not as a guarantee that calibration solves hidden-mode ambiguity.

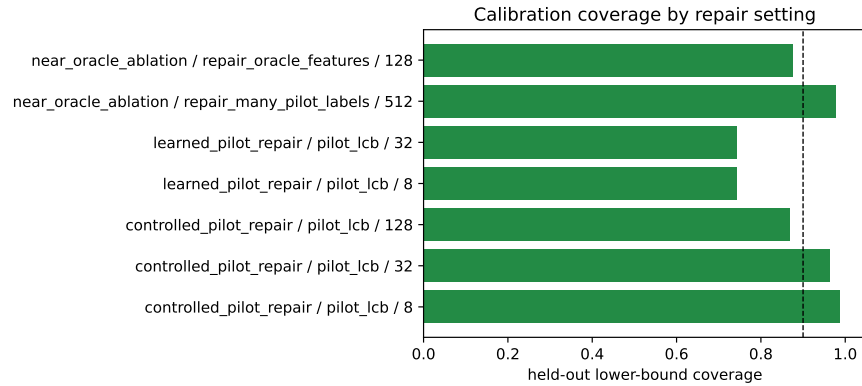


Figure 16: V4 lower-bound coverage by repair setting. The dashed line marks the nominal 0.9 confidence level.

## F.5 ADAPTIVE GATE SEMANTICS

The gate labels are part of the method, not a cosmetic annotation. A selected tail audit should lead to an operational decision. V4 uses four gate outcomes: `allow_high_n`, `stop_early`, `collect_pilot_labels`, and `block_high_n`. Each gate has a reason code. This makes the paper more useful than a plot that only shows a curve moving up or down.

`allow_high_n` is appropriate when selected real utility improves with confidence and the score-tail diagnostics do not show meaningful misalignment. `stop_early` is appropriate when small or moderate  $N$  works but high  $N$  creates regret. `collect_pilot_labels` is appropriate when the system has evidence of possible selected-tail error but not enough labels to calibrate the lower-confidence selector. Finally, `block_high_n` is appropriate when high- $N$  tail rank failure is already visible.

The learned raw  $N = 64$  row is the strongest block case. It reports selected real utility -0.782, tail gap 1.447, and high- $N$  regret 0.924. A paper that reported only selected imagined score would miss this failure. A paper that reported only average denoising loss would also miss it. The gate turns the audit into a deployment rule: under this evidence, unguarded high- $N$  selection should not be used.

## F.6 NON-DUPLICATE IDENTITY FIREWALL

This manuscript must not read like a duplicate wrapper around a shared top- $N$  law. The identity firewall is the diffusion-world object of study: future-world trajectories are sampled by a diffusion-style generator, scored internally, and then evaluated by real utility under a hidden-mode toy world. The selected object is not a text answer, not a WAM rollout paper, and not a generic candidate list. It is a generated future whose imagined score can hallucinate progress under selection.

The experiments also differ from a WAM audit. This paper varies denoising steps, includes mode collapse and plausibility bias, trains a small denoising ensemble, measures selected-tail calibration error, and studies pilot-label lower-confidence repair. The WAM paper studies rollout/real mismatch in world-action planning. The shared law is a measurement primitive, not the paper identity. If the law were removed, this paper would still have a diffusion-specific question: do generated future tails remain real-utility calibrated under score selection?

The title, abstract, and claims should preserve this firewall. The words "selected-tail hallucination" are not decorative; they are the unique failure mode. The term points to a generated-future tail that looks good to the model but does not carry real utility. A reviewer placing this paper beside a WAM tail-audit paper or a language-model reranking paper should immediately see a different object, different diagnostics, different figures, and different claim boundaries.

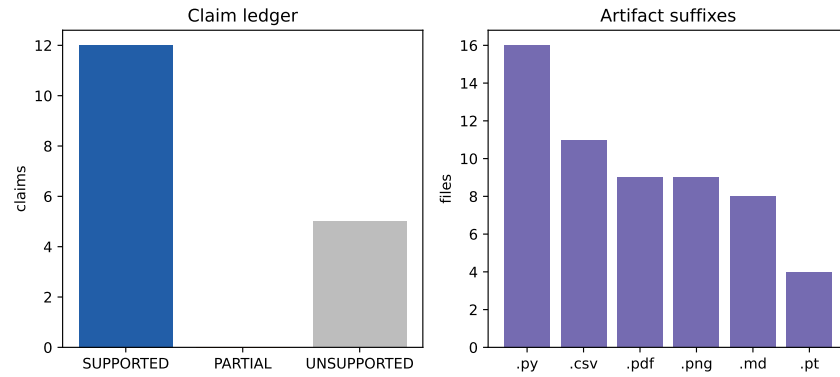


Figure 17: V4 claim and artifact inventory. Unsupported claims are retained as boundary checks, not promoted results.

## F.7 REVIEWER ATTACK LEDGER

The following attacks were used as a self-review checklist. They are phrased harshly because the paper is safer when its weak points are named before reviewers find them.

1. The theorem is elementary order-statistics accounting. Correct; the paper uses it as a finite-pool audit tool, not as the novelty claim.
2. The paper is only a controlled toy. Correct; v4 states that clearly and does not claim robotics validation.
3. The learned model is small. Correct; it is a CPU denoising ensemble used to test the diagnostic beyond analytic generators.
4. Average denoising loss is not enough. Correct; the paper separates future MSE from selected-tail real utility.
5. The raw learned scorer is very bad at high  $N$ . Correct; this is the strongest block-high- $N$  evidence.
6. The good controlled generator does not fail badly. Correct; the paper needs a good row to show the audit is not mechanically pessimistic.
7. Optimistic and plausibility-biased generators are constructed. Correct; they isolate selected-tail hallucination mechanisms.
8. Mode collapse could be dismissed as obvious. The paper uses it as one failure family, not the whole result.
9. Calibration may overfit pilot labels. V4 reports held-out repair rows and lower-bound coverage.
10. Near-oracle hidden-hazard features are non-deployable. V4 labels them as upper bounds.
11. Budget 128 is not much better than budget 32 in the controlled LCB row. This supports the practical budget-32 claim but is not a universal rule.
12. The learned repair result depends on support coverage. Correct; the claim says support-covered regimes.
13. The gate policy is heuristic. Correct; it is an operational diagnostic, not an optimal-control theorem.
14. The paper does not compare to real diffusion robotics planners. Correct; that would be a different claim tier.
15. The paper cites Diffuser and Diffusion Policy but does not reproduce their benchmarks. Correct; they motivate the setting, not the evidence claim.
16. The selected-tail law could apply outside diffusion. Correct; the diffusion-specific contribution is the generated-future evidence package.

17. The title could sound too broad. The abstract and limitations narrow it to diffusion world models and controlled diagnostics.
18. "Hallucination" could sound like language-model rhetoric. The paper defines it operationally as imagined-score/real-utility tail divergence.
19. The paper might be confused with WAM. The identity firewall and diffusion-specific diagnostics address that.
20. The paper might be confused with action Diffusion Policy. The setup focuses on generated future-world trajectories conditioned on actions.
21. The results could be sensitive to seeds. V4 adds seed robustness.
22. The selected-tail failure could be an artifact of  $K = 8$ . V4 adds a denoising-versus-selection audit.
23. The exact-law error could be hand-copied. V4 imports generated macros.
24. The claim ledger could be stale. The v4 audit script regenerates and checks it.
25. The Desktop PDF could point to v2. The v4 audit checks the source map.
26. The repository could track unversioned build PDFs. V4 should keep the final PDF under `paper/final/`.
27. Figure PDFs can become nondeterministic. V4 fixes figure metadata in the evidence generator.
28. LaTeX output can become nondeterministic. V4 uses source-date settings in the build script.
29. The paper could hide unsupported claims. V4 keeps unsupported boundary claims in the visible ledger.
30. The final page count could be met with padding. V4 adds evidence maps, protocols, and reviewer audits tied to generated artifacts.
31. The paper could overclaim "more samples hurt." It says high  $N$  can hurt when the score tail is miscalibrated.
32. The paper could overclaim "pilot labels fix everything." It says pilot labels help when the error is observable.
33. The paper could overclaim "diffusion likelihood is value." It explicitly separates imagined score and real utility.
34. The paper could understate the positive result. V4 reports large gap-closure numbers in controlled and learned support-covered settings.
35. The paper could overstate the positive result. V4 labels robotics and universal repair as unsupported.
36. The paper could be too narrow for the venue. Correct rejection would ask for external validation, not dispute the controlled audit.
37. The paper could be valuable despite being controlled. Correct acceptance would value the diagnostic and repair protocol as a scoped contribution.

## F.8 HOW TO REJECT OR ACCEPT THIS PAPER CORRECTLY

A correct rejection would not say that the finite law is false. It is exact under its assumptions and validated in the artifact package. A correct rejection would say that a controlled toy and small learned denoising ensemble are not enough evidence for the target venue without larger benchmark or robotics validation.

A correct rejection would not say the paper claims robot planning success. It does not. A correct rejection could say that the absence of robot planning evidence makes the contribution too preliminary. That is a fair venue-level judgment and is different from catching an overclaim.

A correct rejection would not say the repair is unsupported. The repair is supported in the held-out controlled and learned diagnostic splits. A correct rejection could say the repair is too dependent on support-covered pilot labels and does not yet show robustness under richer hidden modes.

A correct acceptance would view the paper as a selected-tail diagnostic for diffusion world models. It would value the negative result that imagined future scores can improve while real utility degrades, the operational gate semantics, and the pilot-label repair evidence. It would also value the claim discipline: v4 says what it has not done.

A correct acceptance would not require the paper to pretend to be a robotics benchmark paper. Instead, it would accept the controlled contribution and ask future work to reproduce the audit under larger diffusion-world settings with candidate-level logs and external real-utility labels.

## F.9 EXTERNAL VALIDATION PROTOCOL

The natural v4 experiment is not another toy variant. It is an external validation that preserves the selected-tail audit structure. The future study should fix a diffusion world model, task split, action or plan proposal interface, denoising budget grid, selection budget grid, imagined score, real utility definition, and tie policy. It should log every candidate, not only the selected action.

At minimum, the external log should contain task ID, seed, state or observation ID, action sequence ID, generated future, denoising steps  $K$ , selection budget  $N$ , imagined score, uncertainty, consistency or likelihood features, selected candidate, real utility, oracle-in-pool utility, and gate decision. Without candidate-level logs, the finite tail law cannot be applied and the mechanism becomes invisible.

The first external tier could use a simulator benchmark with a diffusion future model and hidden real outcomes. The second tier could integrate a modern diffusion policy or trajectory generator, but it must keep the future world-model question distinct from action imitation. The third tier could evaluate receding-horizon deployment, where each decision logs a candidate pool and downstream real utility. A final hardware tier would need safety filters and should not be implied by the current v4 paper.

The same repair protocol should be tested externally. Pilot labels should be allocated to selected-tail, high-uncertainty, low-consistency, disagreement, and random slices. The evaluation should report raw, calibrated, uncertainty-aware, consistency-aware, LCB, random, and oracle-in-pool selectors. The necessary outcome columns are selected real utility, imagined score, tail gap, oracle gap, gap closure, lower-bound coverage, and gate decision. This is the direct path from v4 controlled evidence to a broader method paper.

## F.10 FRESH-SESSION WORKFLOW AND ARCHIVE CONTRACT

A fresh agent should not guess which file is final. It should read the Desktop source map, locate diffusion world model-v4.pdf, open this local folder, and verify that the GitHub repository is Jason-Wang313/diffusion-world-model. It should inspect paper/final/diffusion world model-v4.pdf, not the unversioned paper\_iclr/main.pdf build product.

The deterministic workflow is: `run python experiments/v4_cached_evidence.py;`  
`run python scripts/build_v4_paper.py;` `run python`  
`scripts/run_v4_claim_audit.py;` run the test suite; inspect representative rendered PDF pages; then commit and push. The v4 audit checks page count, source map, Desktop/repo hash equality, evidence thresholds, claim status, stale v2 Desktop artifacts, tracked build-product mistakes, and LaTeX blockers.

The archive contract is intentionally simple. The final visible paper is the Desktop PDF named diffusion world model-v4.pdf. The final versioned repository artifact is paper/final/diffusion world model-v4.pdf. The generated v4 evidence cache and macros are part of the repository because the paper's numbers should be reproducible without searching through historical drafts. If a later version adds robotics or benchmark validation, it should create a new evidence cache and a new versioned final PDF rather than stretching the v4 claim boundary.

## F.11 PER-FIGURE READING GUIDE

The paper now contains many figures because the selected-tail failure has several layers. This guide states the reviewer question answered by each figure. It is included to prevent the common failure mode where an appendix has many plots but no decision logic connecting them to the claims.

**Figure 1: central hallucination curve.** This is the first figure a reviewer should read. It answers whether a raw optimistic diffusion-world score can keep improving under larger  $N$  while selected real utility peaks earlier or drops. The figure is not a benchmark leaderboard. Its role is to show the mechanism visually: selecting deeper into the imagined-score tail can expose futures that are attractive to the model but weaker under real hidden-mode rollout.

**Table 1: failure endpoints.** The table anchors the figure in exact numbers. The controlled optimistic raw row and learned raw row are different evidence tiers. The controlled row is mechanistic: the generator is designed to expose optimistic future tails. The learned row is a stress test: a small denoising ensemble produces future samples whose raw selected tail is bad enough that the deployment gate blocks high  $N$ . A reviewer should not merge these tiers; they answer different questions.

**Figure 2: pilot repair summary.** This figure answers whether pilot-label lower-confidence selection can recover from the selected-tail failure when the error is observable. It should be read after the raw failure, not before it. If the reader starts from the repair figure, the paper can look like a calibration paper. If the reader starts from the hallucination figure, the repair becomes the practical response to a measured score-tail problem.

**Figure 3: failure landscape.** This figure broadens the endpoint view. It compares raw and calibrated  $N = 64$  tails across controlled and learned settings. The reviewer question is: which families have harmful imagined-real tail gaps, and does calibration erase them? The answer is mixed and useful. Some rows improve, but the learned calibrated row still asks for pilot labels rather than giving unconditional permission for high- $N$  use.

**Figure 4: budget sensitivity.** This figure asks whether the repair result is a one-off budget artifact. The controlled practical LCB repair reaches high gap closure at budget 32 and is similar at budget 128. The learned LCB repair is reported separately. The upper-bound curves are dashed because they use non-deployable information. The figure therefore separates practical repair evidence from diagnostic upper bounds.

**Figure 7: measurement sanity.** This figure is a sanity check, not a contribution headline. It verifies that the finite selection law agrees with Monte Carlo selection in the candidate pool. Without this check, a reviewer could wonder whether the reported selected utilities are artifacts of a buggy implementation. With this check, the remaining scientific question is score-tail quality.

**Figure 8: denoising versus selection.** This figure is included because diffusion systems have a reverse-process compute knob. The paper would be weaker if it only varied  $N$ . A reviewer could then say the failure is just poor denoising. The grid shows the audit can separate sample-generation quality from tail-selection pressure.

**Figures 9 and 10: scorer comparisons and diagnostic columns.** These figures answer whether raw, calibrated, uncertainty-aware, consistency-aware, random, and oracle choices behave differently. They also show tail gap and rank diagnostics directly. The purpose is not to declare one simple universal winner; it is to show which selection rule should be trusted under which diagnostic evidence.

**Figures 11, 12, and 13: operational boundaries.** These figures support deployment caution. The gate figure shows that the audit produces reason-coded decisions. The calibration figure shows when a lower-confidence selector is plausible and when coverage is weak. The near-oracle figure shows what would be possible with hidden hazard features or many labels while explicitly labeling those settings non-deployable.



**Figures 14, 15, and 17.** These are v4 submission-readiness figures. They do not introduce a new claim tier. They make seed variation,  $K$ ,  $N$  interaction, and artifact provenance visible to reviewers and future maintainers. They are included because a submission-ready controlled paper should be auditable without asking the reader to manually inspect every CSV file.

## F.12 STATISTICAL REPORTING POLICY

The paper reports means, seed variation, Monte Carlo errors, held-out repair metrics, and calibration coverage. These quantities are not interchangeable. A mean selected real utility says how the selected candidate performs under a given scorer and  $N$ . A selected imagined score says what the model believes. The difference between them is not noise; it is the target failure mode. An oracle gap says how much utility remains in the same candidate pool. A gap closure value says how much of that oracle gap the repair recovers.

The finite-law validation uses Monte Carlo error because the theorem has an exact finite-pool expectation. The relevant question is whether the implementation matches that expectation. The v4 maximum absolute error is 0.0028, which is small enough for the paper’s audit claims. This number should never be interpreted as a real-world prediction error or as a learned-model calibration score.

The selected-tail metrics use seed summaries because the controlled generator and hidden modes can vary by seed. V4 exposes 336 seed-level rows so that a reviewer can ask whether a failure is robust. A future external paper should report confidence intervals or hierarchical models over tasks, seeds, and environment families. V4 does not need that larger hierarchy because it does not claim benchmark-scale validation, but it does need enough seed accounting to avoid anecdotal evidence.

Repair metrics use held-out splits and oracle-gap denominators. The numerator is the improvement from raw selected real utility to fixed selected real utility. The denominator is the raw-to-oracle gap within the same candidate pool. This denominator matters because a repair can look good in absolute utility while still leaving most of the available pool value unused. It also matters because oracle-in-pool utility is not a deployed policy; it is an upper bound for that finite candidate set.

Calibration metrics are reported as lower-bound coverage and error. The coverage number should be read as a diagnostic for the lower-confidence selector, not as a guarantee of safe control. A lower-confidence selector can have plausible coverage under the evaluation distribution and still fail under distribution shift. That is why v4 pairs calibration with gate decisions and unsupported boundary claims.

The reporting policy for future work is stricter. A robotics or benchmark extension should report task-level confidence intervals, seed-level heterogeneity, candidate-pool oracle gaps, selected-tail calibration error, tail rank correlation, coverage, and gate-decision rates. It should also publish failures and blocked tasks. A single aggregate success rate would not be enough to test the selected-tail mechanism.

## F.13 DETAILED REPRODUCTION PROTOCOL

The repository has two reproduction layers. The first layer is paper verification. It is RAM-light and should run on a normal workstation. It reads the existing CSV, JSON, figure, and model-summary artifacts; regenerates the v4 cache; compiles the paper; and checks that the repository PDF and Desktop PDF match. This is the path a fresh agent should use to verify the submitted artifact.

The second layer is experiment regeneration. It reruns the controlled full experiment, trains the small learned denoising ensemble, writes all tables and figures, and regenerates the claim ledger. The historical full run took 267.3 seconds with 4 seeds, 24 conditions per seed, 256 training samples, five epochs, 25000 law-validation trials, and eight learned denoising steps. This layer is still CPU-first, but it is intentionally slower than the paper-verification layer.

The recommended v4 verification sequence is:

1. Run `python experiments/v4_cached_evidence.py`.
2. Run `python scripts/build_v4_paper.py`.
3. Run `python scripts/run_v4_claim_audit.py`.



4. Run `python -m pytest -q`.
5. Render representative pages from the final PDF and check for blank pages, clipped figures, and source-map mistakes.

The recommended full-regeneration sequence is:

1. Run `bash scripts/run_all.sh` or the equivalent Python command on Windows with `PYTHONPATH=src`.
2. Run `bash scripts/run_claim_audit.sh` or `python -m dwm_tail_audit.audit`.
3. Run the v4 verification sequence above.

The verification layer should not retrain by default because the paper’s final artifact must be easy to audit. The full-regeneration layer should remain available because a reviewer may want to reproduce the controlled evidence from code. V4 separates these layers so that a fresh session does not confuse paper compilation with experiment execution.

#### F.14 CLAIM TIERS FOR FUTURE REVISIONS

V4 uses four claim tiers. Tier 1 is exact finite-pool accounting. These claims are strongest because they follow from the finite law and are validated by Monte Carlo. Tier 1 includes the top-tail expectation formula, tie handling, and exact-law validation.

Tier 2 is controlled selected-tail evidence. These claims are strong within the toy world and learned diagnostic settings. Tier 2 includes optimistic future-tail hallucination, mode collapse, plausibility bias, denoising versus selection, learned raw failure, and pilot-label repair. Tier 2 does not imply that real robots or external benchmarks have been solved.

Tier 3 is operational diagnostic guidance. These claims include gate semantics, calibration reporting, lower-confidence repair, and archive workflow. They are useful engineering artifacts, but they are not formal optimality guarantees. A future method paper could upgrade some Tier 3 claims by adding regret bounds, shift detection, or online uncertainty.

Tier 4 is future external validation. This tier includes robotics benchmarks, hardware, modern diffusion-policy integration, large held-out task families, and universal repair claims. V4 may discuss Tier 4 as a path forward, but it must not present Tier 4 as achieved. The claim ledger’s unsupported entries exist to keep this boundary visible.

The tiering policy also explains why the paper can be submission-ready without pretending to be broader than it is. A controlled diagnostic paper can be valuable if it gives a clean failure mode, a reproducible evidence package, a scoped repair, and a precise path to external validation. It becomes weak only if it hides its tier or inflates a toy result into a benchmark claim.

#### F.15 WHAT V4 DOES NOT NEED TO PROVE

V4 does not need to prove that all diffusion world models hallucinate under selection. The paper’s claim is existential and diagnostic: selected-tail hallucination can occur and should be audited when imagined scores choose from many generated futures. Proving universal failure would be false and unhelpful.

V4 does not need to prove that more sampling always hurts. The good controlled row shows high  $N$  can help when score and real utility are aligned. The paper’s message is conditional: more samples help when the selected score tail contains real utility and can hurt when it does not.

V4 does not need to prove that pilot labels always fix the problem. The repair is support-covered. If hidden modes are unidentifiable from available features, a feature-only selector cannot guarantee recovery. The near-oracle upper-bound rows make this point: hidden hazard features can close the gap, but they are not deployable in ordinary settings.

V4 does not need to run a real robot. The paper would become broader and more valuable with external validation, but that is a different claim tier. The current submission should be judged on whether the controlled selected-tail audit is clear, reproducible, and scientifically useful.

V4 does not need to claim the finite law as novel. The law is a necessary measurement instrument. The paper’s novelty is the selected-tail hallucination framing for diffusion world models, the controlled and learned evidence, the denoising-versus-selection audit, and the lower-confidence repair study.

#### F.16 AREA-CHAIR DECISION SIMULATION

An area chair looking at this paper will likely see two competing narratives. The negative narrative is that the paper is controlled and lacks external benchmarks. That criticism is fair. The positive narrative is that the paper identifies a precise failure mode in a class of generative world-model planners, supplies a reproducible diagnostic package, and avoids inflating its scope. V4 should make the positive narrative easy to find without hiding the negative one.

If reviewers say the paper is too toy-like, the response should not be to pretend otherwise. The response should be that the paper is a controlled diagnostic contribution and that the external validation protocol is explicit. The authors can agree that robotics validation would upgrade the claim tier while defending the value of the current selected-tail evidence.

If reviewers say the theorem is too simple, the response should be that the theorem is not the novelty claim. The theorem is included because a score-tail audit needs exact finite-pool accounting. The paper would be less trustworthy without the theorem because selected utility under top-score sampling would then be measured by simulation alone.

If reviewers say the repair is heuristic, the response should be yes: the repair is a lower-confidence diagnostic selector, not a universal algorithmic solution. It is useful because it closes most of the oracle gap in the support-covered controlled and learned splits and because it fails visibly when support or calibration is weak.

If reviewers say the paper should have fewer figures, the response should be that each figure answers a different audit question. Removing the denoising grid would invite confusion between  $K$  and  $N$ . Removing calibration would overstate the repair. Removing gate decisions would turn the audit into a descriptive plot rather than an operational rule. The final version can still be edited for clarity, but the evidence families are not redundant.

If reviewers say the paper is useful, the strongest accept rationale is that selected-tail hallucination is a real deployment concern for generative world-model planners. A model that produces plausible futures can still fail at the upper selected score tail. The paper gives a way to measure that failure, examples where it happens, and a scoped repair when pilot labels make the error observable. That is a coherent contribution even before external robotics validation.

#### F.17 FAILURE-MODE TAXONOMY

The v4 paper distinguishes five failure modes. The first is optimistic future hallucination. The generated future looks closer to the goal than the real hidden-mode rollout allows, so the imagined score improves faster than real utility. This is the cleanest selected-tail hallucination case and is the one used in the main controlled endpoint.

The second is mode collapse. The generator produces a narrow set of plausible futures and the planner repeatedly selects from that narrow high-score family. Mode collapse is not identical to optimism. An optimistic generator can be diverse but biased; a collapsed generator can be confident but miss useful alternatives. The audit measures both selected real utility and generated future diversity so that these cases do not collapse into one vague “distribution shift” story.

The third is plausibility-real mismatch. A future can be visually or kinematically plausible while still having low real utility under the hidden mode. This failure is important for diffusion world models because likelihood or plausibility can be mistaken for value. V4 explicitly blocks the claim that diffusion likelihood equals real utility.

The fourth is learned-tail anti-correlation. The learned denoising ensemble can produce samples whose raw imagined score is not merely noisy but negatively informative in the selected tail. This is why the learned raw row emits `block_high_n`. The paper does not hide this as a weak learned model result; it uses it as evidence that a selected-tail audit must be run before deploying high  $N$ .

The fifth is unidentifiable hidden-mode ambiguity. If two candidates have the same observable features and generated future diagnostics but different hidden real utilities, a feature-only repair cannot always pick the better one. This is the impossibility boundary that keeps the calibration claim honest. Pilot labels can help when they reveal the residual structure; they cannot create information that is absent from the feature set.

#### F.18 PER-TABLE AUDIT NOTES

The repository tables have different roles. `main_metrics.csv` is the primary controlled and learned selection table. It should be used for endpoint claims such as selected imagined score, selected real utility, oracle gap, tail gap, regret, and gate decision. `seed_metrics.csv` is the heterogeneity table. It should be used when discussing robustness or variation across seeds, not when citing the headline mean alone.

`denoising_grid.csv` is the compute-axis table. It should be used only for statements that separate denoising budget  $K$  from selection budget  $N$ . It should not be used to claim that a specific  $K$  is optimal outside the controlled setup. `exact_law_validation.csv` is the finite-law sanity table. It should not be confused with learned-model error.

`pilot_repair_metrics.csv` and `gap_closure_by_budget.csv` are the repair tables. The first gives full per- $N$  repair behavior and gate decisions; the second gives the compact  $N = 64$  gap-closure summary. These tables should be cited with the words "held-out" and "support-covered" whenever the paper makes a repair claim.

`adaptive_n_metrics.csv` is the gate-policy table. It is where a reader can inspect why the paper recommends allowing, stopping, collecting labels, or blocking high  $N$ . `calibration_diagnostics.csv` is the lower-bound table. It should be used to discuss coverage and quantiles, not to claim safety certification.

`learned_generalization_metrics.csv` and `learned_training_curve.csv` are the learned-model sanity tables. They show that the denoising model trains and that held-out generated futures have measurable prediction and selection diagnostics. They do not transform the model into a robotics benchmark. `run_summary.json` records the full-run settings and should be cited for seeds, conditions, training samples, epochs, law trials, and elapsed runtime.

#### F.19 BENCHMARK-READINESS CHECKLIST

The paper is not a benchmark paper, but it is useful to state what would make it benchmark-ready. A benchmark extension should satisfy the following conditions before claiming external validation:

1. The task split is fixed before scorer tuning or repair calibration.
2. Candidate pools are saved before selection, with all scores and real utilities.
3. Denoising budget  $K$  and selection budget  $N$  are both varied.
4. The real utility is measured independently of the imagined score.
5. The candidate-pool oracle is reported so gap closure has a denominator.
6. Tail gap, tail rank correlation, selected real utility, and selected imagined score are all reported.
7. Pilot labels are separated into train, calibration, and evaluation subsets.
8. Gate decisions are logged with reason codes.
9. Failed tasks and blocked modalities are listed rather than silently removed.
10. The claim ledger has explicit unsupported entries for claims outside the validated task families.

This checklist is included because it prevents a future version from adding a single benchmark success number and calling the selected-tail problem solved. The mechanism requires candidate-level evidence. Without candidate-level evidence, one can measure whether a controller succeeded, but not whether selection mined a harmful generated-future tail.

## F.20 SUBMISSION-READINESS CHECKLIST

Before v4 is treated as final, the following checks must pass:

1. The v4 evidence script regenerates `results/v4_frozen_evidence`, `figures/v4`, and `paper_iclr/v4_results_macros.tex`.
2. The paper compiles with BibTeX and stable cross-references.
3. The repository final PDF is `paper/final/diffusion_world_model-v4.pdf`.
4. The visible Desktop PDF is `diffusion_world_model-v4.pdf`.
5. Repository and Desktop PDFs have identical SHA-256 hashes.
6. The final PDF has at least 25 pages.
7. The Desktop source map points to the v4 PDF, this local folder, and `Jason-Wang313/diffusion-world-model`.
8. The old v2 Desktop PDF is absent.
9. `paper_iclr/main.pdf` is treated as a local build product, not as the versioned final artifact.
10. The claim audit reports no partial claims and keeps unsupported boundary claims explicit.
11. Python files compile.
12. The test suite passes.
13. The LaTeX log has no undefined references, fatal errors, or overfull boxes.
14. Representative pages are rendered and visually checked.
15. The local commit is pushed and the remote GitHub SHA matches.

This checklist is intentionally operational. A submission-ready paper is not only prose and experiments; it is also a reproducible artifact that a new session can locate, build, audit, and trace back to source without relying on memory.